# AICE3001/AICE6002: Mathematics of Machine Learning

Subsidiary Notes — Lecture 26: Information Theory

Adam Prugel-Bennett

November 2, 2026

## 1 Keywords

- Conditional entropy, joint entropy, mutual information, channel capacity, KL-divergence, variational inference, ELBO, cross-entropy, minimum description length, variational auto-encoders

## 2 Main Points

### 2.1 Communicating Down a Noisy Channel

- Information theory studies sending a message $X$, drawn with probability $\mathbb{P}[X]$, down a noisy channel that delivers $Y$ with probability $\mathbb{P}[Y \mid X]$
- Having received $Y = y$, our remaining uncertainty about what was sent is

$$H_{X|Y=y} = -\sum_{x \in \mathcal{X}} \mathbb{P}[X = x \mid Y = y] \log(\mathbb{P}[X = x \mid Y = y])$$

- Averaging over what we might receive gives the *conditional entropy*

$$H_{X|Y} = \sum_{y \in \mathcal{Y}} \mathbb{P}[Y = y]\ H_{X|Y=y} = -\sum_{x,y} \mathbb{P}[X = x, Y = y] \log(\mathbb{P}[X = x \mid Y = y])$$

### 2.2 Joint Entropy

- The *joint entropy* is the entropy of the pair,

$$H_{X,Y} = -\sum_{x,y} \mathbb{P}[X = x, Y = y] \log(\mathbb{P}[X = x, Y = y])$$

- If $X$ and $Y$ are independent then $H_{X,Y} = H_X + H_Y$ (see the previous lecture); if they are dependent, $H_{X,Y} < H_X + H_Y$
- Taking logs of $\mathbb{P}[X, Y] = \mathbb{P}[Y \mid X]\ \mathbb{P}[X] = \mathbb{P}[X \mid Y]\ \mathbb{P}[Y]$ and averaging gives the *chain rule* for entropy

$$H_{X,Y} = H_X + H_{Y|X} = H_Y + H_{X|Y}$$

and hence $H_X - H_{X|Y} = H_Y - H_{Y|X}$

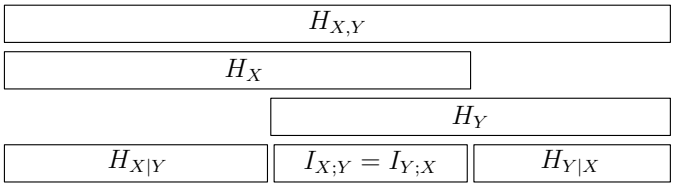
## 2.3 Mutual Information

- Before receiving anything our uncertainty about $X$ is $H_X$. Shannon defined the *mutual information* as the expected reduction in that uncertainty when we receive the message:

$$I_{X;Y} = H_X - H_{X|Y}$$

- By the chain rule result above, $I_{X;Y} = I_{Y;X}$: $Y$ tells us as much about $X$ as $X$ tells us about $Y$

- The relationships between these quantities are summarised in the diagram below




- The *capacity* of a channel is the largest information gain we can achieve by choosing the distribution of messages,

$$C = \max_{\mathbb{P}[X]} I_{X;Y}$$

- Mutual information is widely used in machine learning; for example, two sets of images can be aligned by choosing the non-linear transformation that maximises the (normalised) mutual information between them

## 2.4 The Binary Symmetric Channel

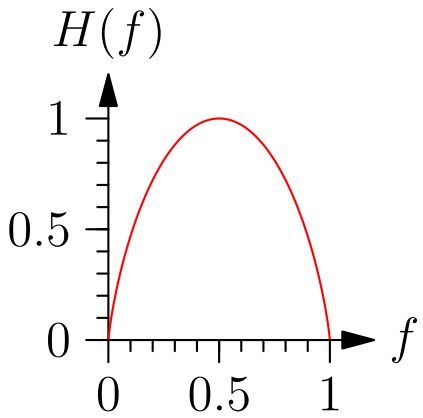
- The simplest noisy channel sends bits, each flipped independently with probability $f$:

$$\mathbb{P}[Y = 1 \mid X = 0] = \mathbb{P}[Y = 0 \mid X = 1] = f$$

- With equally likely inputs, an elementary calculation gives, per bit,

$$H_{X_i|Y_i} = -(1-f)\log_2(1-f) - f\log_2(f) = H(f)$$

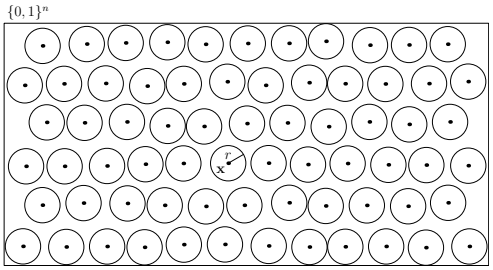
  and for a message of $n$ bits $H_{X|Y} = n\,H(f)$




- Since $H_X = n$ bits for uniform inputs, $I_{X;Y} = n\,(1 - H(f))$. Uniform inputs are optimal here, so the capacity is $1 - H(f)$ bits per bit sent

## 2.5 Error Correcting Codes and Shannon's Bound

- To communicate reliably we use an error correcting code: choose code words far apart in the space of binary strings, so that a corrupted code word is still closer to the original than to any other




- A *Hamming ball* $B(\boldsymbol{x}, r)$ is the set of strings that differ from $\boldsymbol{x}$ in at most $r$ positions
- A string of length $n$ suffers about $n\,f$ errors, and for large $n$ almost never more than $(f+\epsilon)\,n$. So if each code word owns a ball of radius $r = (f+\epsilon)\,n$, and the balls don't overlap, errors can be corrected
- By Stirling's approximation a ball of radius $f\,n$ contains $\binom{n}{fn} \approx 2^{nH(f)}$ strings, so at most

$$\frac{2^n}{2^{nH(f)}} = 2^{n(1-H(f))}$$

  non-overlapping balls fit. We can therefore send at most $n\,(1-H(f))$ bits of information in $n$ bits: the rate cannot exceed the capacity
- Shannon also showed that choosing $2^{nI_{X;Y}}$ code words at random keeps them far enough apart with high probability, so this rate is *achievable*
- **Example.** For $f = 0.1$, $H(0.1) \approx 0.469$ bits, so the capacity is $C = 1 - 0.469 \approx 0.531$ bits. Reliable communication needs codes about $1/0.531 \approx 1.9$ times as long as the message—just under twice as long—for a channel that corrupts 10% of bits
- Random codes cannot be decoded efficiently, so Shannon's result told us the bound was achievable without telling us how. Practical codes that come close to it (such as turbo and LDPC codes) took decades more to find

## 2.6 KL-Divergence

- The *Kullback–Leibler divergence* is

$$\mathrm{KL}\,(p \,\|\, q) = \mathbb{E}_{X\sim p}\left[\log\left(\frac{p(X)}{q(X)}\right)\right] = \mathbb{E}_{X\sim p}[-\log(q(X))] - H_p$$

- $-\log(q(x))$ is the code length for $x$ in the optimal code for the distribution $q$. So

$$\mathbb{E}_{X\sim p}[-\log(q(X))]$$

  is the expected message length when data from $p$ is coded as if it came from $q$, and $\mathrm{KL}\,(p \,\|\, q)$ is the expected *extra* length paid for using the wrong distribution
- It is also called the *relative entropy*. It is never negative, is zero only when $p = q$, and is not symmetric: $\mathrm{KL}\,(p \,\|\, q) \neq \mathrm{KL}\,(q \,\|\, p)$ in general

## 2.7 Cross-Entropy Loss

- A classifier gives probabilities $q(y \mid \boldsymbol{x})$ and is usually trained by minimising the *cross-entropy*

$$\mathcal{L} = -\frac{1}{m}\sum_{k=1}^{m} \log(q(y_k \mid \boldsymbol{x}_k))$$

  which is the average code length for the labels using the model's probabilities

- If, at a given $\boldsymbol{x}$, the labels really come from $p(y \mid \boldsymbol{x})$,

$$\mathbb{E}_{Y\sim p}[-\log(q(Y \mid \boldsymbol{x}))] = H_p + \mathrm{KL}\,(p \,\|\, q)$$

  (add and subtract $\log p$). The entropy $H_p$ does not depend on the model, so minimising the cross-entropy minimises the KL-divergence from the data to the model

## 2.8 Variational Approximation

- In Bayesian inference the posterior $f(\boldsymbol{\theta} \mid \mathcal{D})$ is often too complicated to write down, which is why we used MCMC

- The *variational approximation* instead fits a simpler distribution, typically factorised,

$$f(\boldsymbol{\theta} \mid \mathcal{D}) \approx g(\boldsymbol{\theta} \mid \boldsymbol{\phi}) = \prod_i g(\theta_i \mid \phi_i)$$

  by maximising the *evidence lower bound* (ELBO; it is minus the *variational free energy* of physics)

$$\Phi(\boldsymbol{\phi}) = -\int g(\boldsymbol{\theta} \mid \boldsymbol{\phi}) \, \log\left(\frac{g(\boldsymbol{\theta} \mid \boldsymbol{\phi})}{f(\boldsymbol{\theta}, \mathcal{D})}\right) \mathrm{d}\boldsymbol{\theta}$$

## 2.9 The Evidence Lower Bound (ELBO)

- Writing $f(\boldsymbol{\theta}, \mathcal{D}) = f(\boldsymbol{\theta} \mid \mathcal{D})\, f(\mathcal{D})$,

$$\begin{aligned}\Phi(\boldsymbol{\phi}) &= -\int g(\boldsymbol{\theta} \mid \boldsymbol{\phi}) \left(\log\left(\frac{g(\boldsymbol{\theta} \mid \boldsymbol{\phi})}{f(\boldsymbol{\theta} \mid \mathcal{D})}\right) - \log(f(\mathcal{D}))\right) \mathrm{d}\boldsymbol{\theta} \\ &= \log(f(\mathcal{D})) - \mathrm{KL}\,(g(\boldsymbol{\theta} \mid \boldsymbol{\phi}) \,\|\, f(\boldsymbol{\theta} \mid \mathcal{D}))\end{aligned}$$

- Because the KL-divergence is non-negative, $\Phi(\boldsymbol{\phi}) \le \log(f(\mathcal{D}))$: $\Phi$ is a lower bound on the log evidence, hence the name ELBO

- Maximising $\Phi$ minimises the KL-divergence, making $g$ close to the true posterior and $\Phi \approx \log(f(\mathcal{D}))$

- Equivalently $\Phi(\boldsymbol{\phi}) = L_g(\boldsymbol{\phi}) + H_g(\boldsymbol{\phi})$ with

$$L_g(\boldsymbol{\phi}) = \int g(\boldsymbol{\theta} \mid \boldsymbol{\phi}) \, \log(f(\mathcal{D} \mid \boldsymbol{\theta})\, f(\boldsymbol{\theta})) \, \mathrm{d}\boldsymbol{\theta} \qquad H_g(\boldsymbol{\phi}) = -\int g(\boldsymbol{\theta} \mid \boldsymbol{\phi}) \, \log(g(\boldsymbol{\theta} \mid \boldsymbol{\phi})) \, \mathrm{d}\boldsymbol{\theta}$$

  The first term rewards putting $g$ where the joint density is large (compare MAP); the entropy term rewards keeping $g$ as spread out as possible

- Variational methods are usually much faster than MCMC and can be very accurate, but the quality is not guaranteed. MCMC is a controlled approximation that improves with more iterations; its slowness is rarely a problem on modern computers

- Minimising the divergence the other way round, $\mathrm{KL}\,(f \,\|\, g)$, leads to a different family of methods (expectation propagation)

## 2.10 Minimum Description Length

- Outside the Bayesian framework, choosing between models of different complexity is hard. A validation set is the usual answer, but is not always possible
- A principled alternative: prefer the model that *compresses the data most*. If a model compresses the data it has captured real structure
- **Alice and Bob.** Alice has data $\mathcal{D} = \{(\boldsymbol{x}_i, y_i) \mid i = 1, \ldots, m\}$; Bob has only the inputs $\boldsymbol{x}_i$. Both have a model $\hat{f}(\boldsymbol{x} \mid \boldsymbol{\theta})$. Rather than sending all the $y_i$, Alice sends the parameters $\boldsymbol{\theta}$ and the errors $\delta_i = y_i - \hat{f}(\boldsymbol{x}_i \mid \boldsymbol{\theta})$
- If the errors have density $p_\delta$, sending one error to accuracy $\Delta$ costs $-\log(p_\delta(\delta_i)\,\Delta)$. The *description length* is

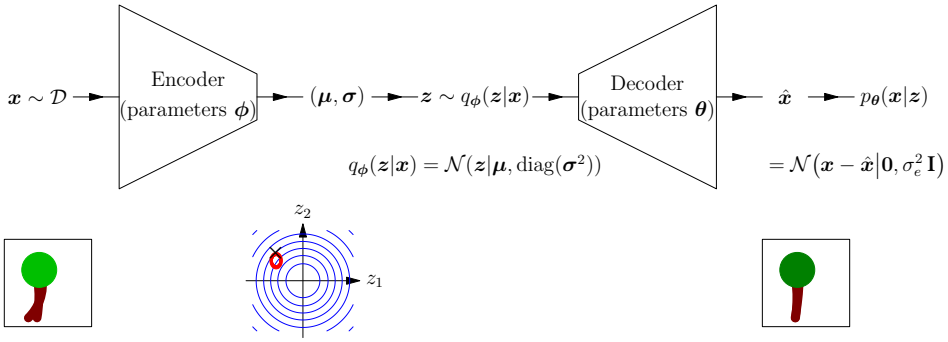
$$L = \sum_{k=1}^{n} \ell(\theta_k) - \sum_{i=1}^{m} \left( \log\Big( p_\delta\Big( y_i - \hat{f}(\boldsymbol{x}_i \mid \boldsymbol{\theta}) \Big) \Big) + \log(\Delta) \right)$$

  where $\ell(\theta_k)$ is the cost of sending parameter $\theta_k$, to whatever precision is worthwhile
- The *minimum description length* (MDL) principle chooses the model with the smallest $L$. The $-m \log(\Delta)$ term is the same for every model and does not affect the choice
- For Gaussian errors $-\log(p_\delta(\delta))$ is the squared error plus a constant, so MDL is least squares plus a penalty for model complexity
- MDL trades accuracy against complexity and is often the only principled method available. It can be fiddly because we must decide how precisely to store the parameters, though good models often survive surprisingly low precision

## 2.11 Variational Auto-Encoders

- A variational auto-encoder (VAE) encodes an input $\boldsymbol{x}$ into a distribution $q_{\boldsymbol{\theta}}(\boldsymbol{z} \mid \boldsymbol{x})$ over a latent code $\boldsymbol{z}$ and decodes $\boldsymbol{z}$ back to a reconstruction. Its loss is

$$\mathcal{L} = \mathbb{E}_{\boldsymbol{x} \sim \mathcal{D}}[\mathrm{KL}\left(q_{\boldsymbol{\theta}}(\boldsymbol{z} \mid \boldsymbol{x}) \,\|\, \mathcal{N}(\mathbf{0}, \mathbf{I})\right) - \log(p_{\boldsymbol{\theta}}(\boldsymbol{x} \mid \boldsymbol{z}(\boldsymbol{x})))]$$




- The original paper derived this as a variational lower bound on the likelihood of the data: the loss is minus an ELBO, with $q_{\boldsymbol{\theta}}(\boldsymbol{z} \mid \boldsymbol{x})$ as the variational distribution. That derivation is hard to follow. The loss has a much more natural reading as a description length:
  - Alice encodes each image into a latent code $q_{\boldsymbol{\theta}}(\boldsymbol{z} \mid \boldsymbol{x})$ and sends it, at a cost of

$$\mathrm{KL}\left(q_{\boldsymbol{\theta}}(\boldsymbol{z} \mid \boldsymbol{x}) \,\|\, \mathcal{N}(\mathbf{0}, \mathbf{I})\right)$$

    (strictly, this is the cost with "bits-back" coding, Hinton and van Camp 1993: the randomness in the sampled code can carry other information, which is refunded)

- she also sends the errors needed to repair Bob's reconstruction, at a cost of

$$-\log(p_{\boldsymbol{\theta}}(\boldsymbol{x} \mid \boldsymbol{z}(\boldsymbol{x})))$$

- Minimising the loss minimises the total description length. The clever part is that the network also chooses how precisely to send the code, trading code cost against repair cost

# 3 Exercises

## 3.1 The binary symmetric channel

- Verify that with equally likely inputs $H_{X_i|Y_i} = H(f)$, and hence that $I_{X_i;Y_i} = 1 - H(f)$ bits. Evaluate the capacity for $f = 0.1$ and $f = 0.5$, and explain the second answer

## 3.2 Mutual information from entropies

- Use the chain rule for entropy to show that $I_{X;Y} = H_X + H_Y - H_{X,Y}$

## 3.3 The KL-divergence is not symmetric

- For two coins with heads probabilities $p = 0.5$ and $q = 0.9$, compute $\mathrm{KL}\,(p \,\|\, q)$ and $\mathrm{KL}\,(q \,\|\, p)$ in bits

## 3.4 KL-divergence between normals (harder)

- Show that

$$\mathrm{KL}\left(\mathcal{N}(\mu_1, \sigma_1^2) \,\|\, \mathcal{N}(\mu_2, \sigma_2^2)\right) = \log\left(\frac{\sigma_2}{\sigma_1}\right) + \frac{\sigma_1^2 + (\mu_1 - \mu_2)^2}{2\,\sigma_2^2} - \frac{1}{2}$$

  in nats

# 4 Answers

## 4.1 The binary symmetric channel

- With uniform inputs and symmetric noise, Bayes' rule gives $\mathbb{P}[X = y \mid Y = y] = 1 - f$ and $\mathbb{P}[X \neq y \mid Y = y] = f$ whichever $y$ is received, so $H_{X_i|Y=y} = H(f)$ for both $y$, and the average is $H(f)$
- $H_{X_i} = 1$ bit, so $I_{X_i;Y_i} = 1 - H(f)$
- $f = 0.1$: $C \approx 1 - 0.469 = 0.531$ bits. $f = 0.5$: $H(0.5) = 1$, so $C = 0$. When each bit is flipped with probability one half the output is independent of the input, and nothing gets through

## 4.2 Mutual information from entropies

- From $H_{X,Y} = H_Y + H_{X|Y}$ we have $H_{X|Y} = H_{X,Y} - H_Y$, so

$$I_{X;Y} = H_X - H_{X|Y} = H_X + H_Y - H_{X,Y}$$

## 4.3 The KL-divergence is not symmetric

- $\mathrm{KL}\,(p \,\|\, q) = 0.5 \log_2(0.5/0.9) + 0.5 \log_2(0.5/0.1) \approx 0.737$ bits
- $\mathrm{KL}\,(q \,\|\, p) = 0.9 \log_2(0.9/0.5) + 0.1 \log_2(0.1/0.5) \approx 0.531$ bits

### 4.4 KL-divergence between normals (harder)

- With $p = \mathcal{N}(\mu_1, \sigma_1^2)$ and $q = \mathcal{N}(\mu_2, \sigma_2^2)$,

$$\log\left(\frac{p(x)}{q(x)}\right) = \log\left(\frac{\sigma_2}{\sigma_1}\right) - \frac{(x-\mu_1)^2}{2\,\sigma_1^2} + \frac{(x-\mu_2)^2}{2\,\sigma_2^2}$$

- Take the expectation over $X \sim p$. $\mathbb{E}\left[(X-\mu_1)^2\right] = \sigma_1^2$, and writing $X - \mu_2 = (X-\mu_1) + (\mu_1 - \mu_2)$, $\mathbb{E}\left[(X-\mu_2)^2\right] = \sigma_1^2 + (\mu_1-\mu_2)^2$ because the cross term has zero mean

- Hence

$$\mathrm{KL}\left(p \,\|\, q\right) = \log\left(\frac{\sigma_2}{\sigma_1}\right) - \frac{1}{2} + \frac{\sigma_1^2 + (\mu_1-\mu_2)^2}{2\,\sigma_2^2}$$